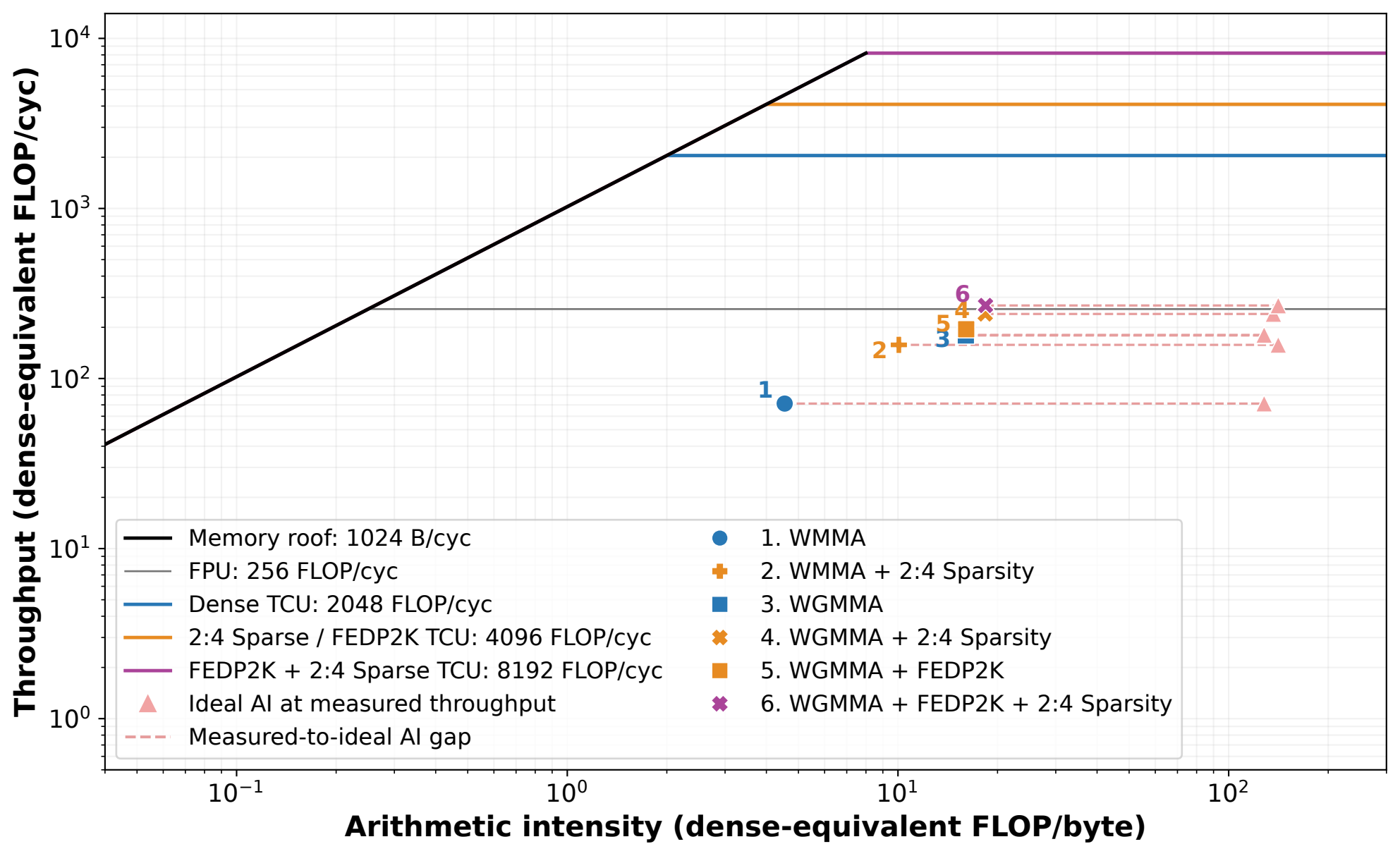
Throughput (dense-equivalent FLOP/cyc)
10^4
10^3
10^2
10^1
10^0
10^{-1}
10^0
10^1
10^2
Arithmetic intensity (dense-equivalent FLOP/byte)
1
2
3
4
5
6
Memory roof: 1024 B/cyc
FPU: 256 FLOP/cyc
Dense TCU: 2048 FLOP/cyc
2:4 Sparse / FEDP2K TCU: 4096 FLOP/cyc
FEDP2K + 2:4 Sparse TCU: 8192 FLOP/cyc
Ideal AI at measured throughput
Measured-to-ideal AI gap
1. WMMA
2. WMMA + 2:4 Sparsity
3. WGMMA
4. WGMMA + 2:4 Sparsity
5. WGMMA + FEDP2K
6. WGMMA + FEDP2K + 2:4 Sparsity